

Kyle Fernandez

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EDUCATION

York University

Bachelor of Science Specialized Honours in Computer Science

Toronto, ON, M3N 2P5

Expected June 2028

EXPERIENCE

Full-Stack Developer (Remote Freelance) | *Beyond Education Academy*

Mar 2025 – Present

- Built a full-stack SAT, AP & IB Exam College Board test simulator using Next.js, React, and Tailwind with responsive layouts for mobile and desktop, **serving 200+ students**
- Integrated Supabase Auth & PostgreSQL for real-time answer syncing, timer tracking, and cross-device session resumption
- Optimized application performance** through server-side rendering, efficient database queries, and reduced client-side bundle size, improving load times across devices, **decreasing response time from 30 seconds → 5 seconds**
- Built custom SQL reporting pipelines tracking domain mastery, difficulty-based accuracy, and exam fatigue patterns across 200+ students, **contributing to individual score gains of up to 400 points in 90 days**

AI Automation Developer (Remote Freelance) | *Paul's Academy*

Dec 2024 – Mar 2025

- Built an AI-driven exam automation tool that **reduced exam creation time by 97% (3 hours → under 5 minutes)**
- Developed a React front-end and JavaScript back-end to parse PDF exams and auto-populate Google Sheets via the Drive & Sheets API (Google Cloud)
- Fine-tuned a FLAN-T5 model with rule-based classification logic in Python, achieving **99.1% question type accuracy** across **1,000+ SAT exam questions**
- Configured multi-LLM pipelines using OpenAI models to generate differentiated exam content based on question classification, difficulty tier, and subject matter

PROJECTS

[Automated Vulnerability Assessment & Penetration Testing](#) | *Python, Nmap, Metasploit, NVD/CIRCL APIs, ReportLab*

May 2026

- Built a **vulnerability scanning** pipeline using Nmap's Vulners engine, cross-referencing CVEs against NVD/CIRCL APIs for CVSS scores and remediation data
- Automated penetration testing** by querying msfconsole for each CVE and executing matching Metasploit modules against a homelab to validate exploitability
- Generated PDF risk-assessment reports via ReportLab with risk-distribution summaries, top-5 critical CVE vector analysis, and a full CVSS-sorted vulnerability table with remediation guidance

[YorkU Event Hub - Full-Stack Developer](#) | *React, FastAPI, Supabase (PostgreSQL), Jira, Git*

Sep 2025

- Built a centralized event platform giving students real-time access to campus & club activities, backed by a normalized PostgreSQL database
- Built a Python web scraper and Discord bot to automate event data extraction from **50+ club channels in under 30 seconds**
- Led and collaborated in an Agile-based** team of six using Jira for sprint planning and Git for branching/rebasing workflows

[American Sign Language Learning App \(UNHack Hackathon\)](#) | *React, Python, OpenCV, MediaPipe, Keras*

December 2024

- Built an accessible "Duolingo-style" web app to teach American Sign Language, with a React frontend connected to a Python backend
- Used MediaPipe and OpenCV for real-time hand-landmark tracking and trained a Keras model to classify signs, providing instant feedback on sign accuracy

TECHNICAL SKILLS & CERTIFICATIONS

- Certifications & Related Coursework:** CompTIA Security+, Computer Networks (OSI Model)
- Languages & Tools:** Java, C, Python, JavaScript/TypeScript, Bash, SQL, Swift, Tableau, Power BI, Excel (Macros)
- Frameworks, Libraries & Databases:** React.js, Next.js, Three.js, Node.js, MySQL, Supabase, AWS, Azure, Google Cloud
- Management Platforms:** Jira, Trello, Git, Notion, Slack, Teams, SIEM Splunk